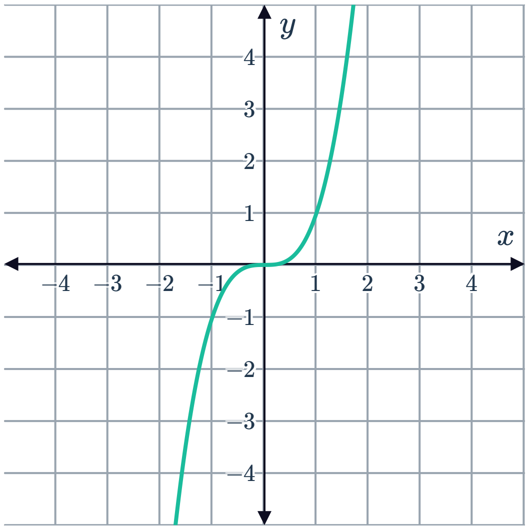
The shape of a cubic

1 Consider the graph of $y = x^3$.

a As x becomes larger in the positive direction (ie x approaches infinity), what happens to the corresponding y-values?

b As x becomes larger in the negative direction (ie x approaches negative infinity), what happens to the corresponding y-values?

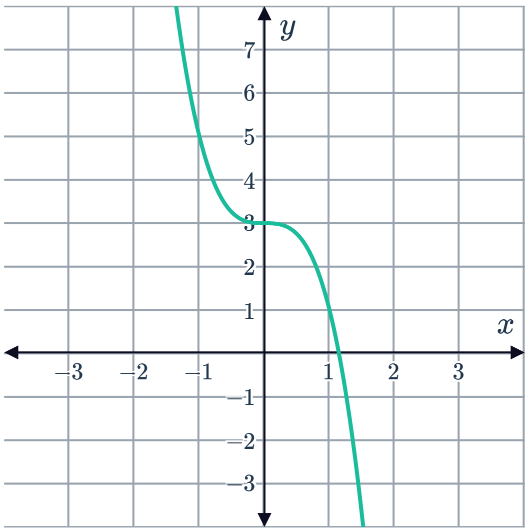


2 Consider the given graph of a cubic function.

a Determine whether the cubic is positive or negative.

b State the coordinates of the y-intercept.

c State the equation of the function.

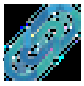


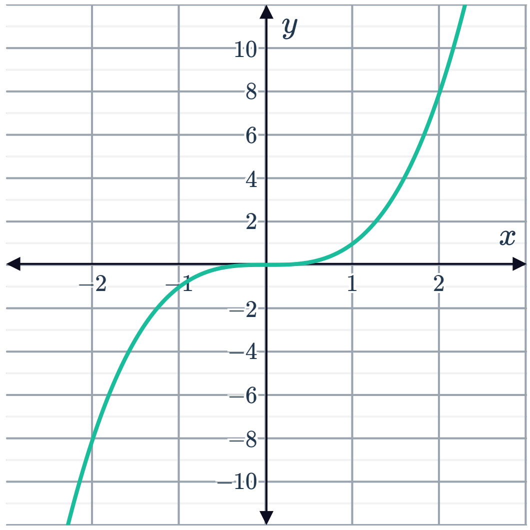
3 Consider the cubic function $y = \frac{1}{2}x^3 + x$.

a Determine whether the cubic is positive or negative.

b Sketch the graph of $y = \frac{1}{2}x^3 + x$.

c State the coordinates of the x-intercept.

4 Consider the graph of the function $y = x^3$ 
Determine the point where the curve changes from being concave down to being concave up.
This is called the point of inflection.

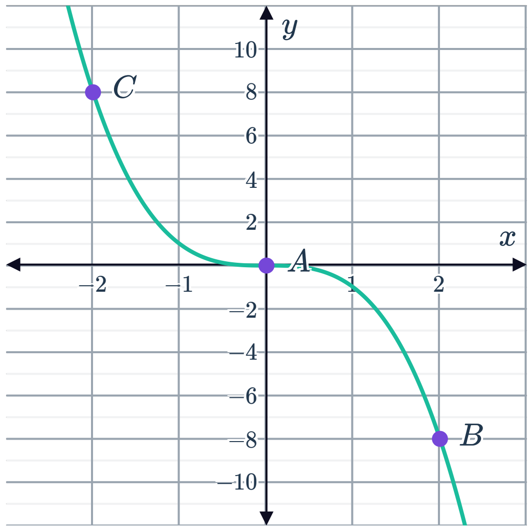


5 Consider the graph of the function $y = -x^3$.
Out of the points A, B and C:

a At which point is the curve concave up?

b Which is the point of inflection?

c At which point is the curve concave down?



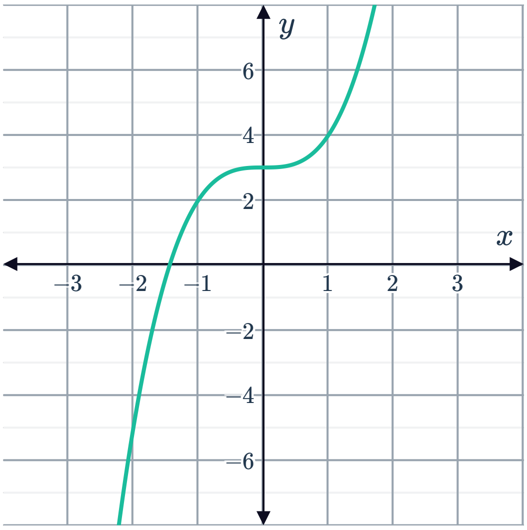
6 For each of the following functions:

i For what values of x is the cubic concave up?

ii For what values of x is the cubic concave down?

iii State the coordinates of the point of inflection.

a



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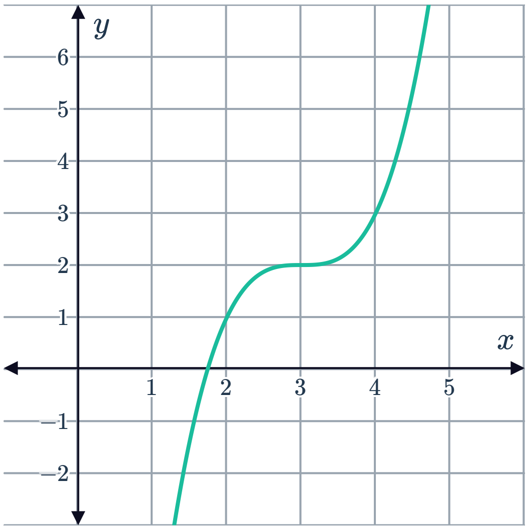
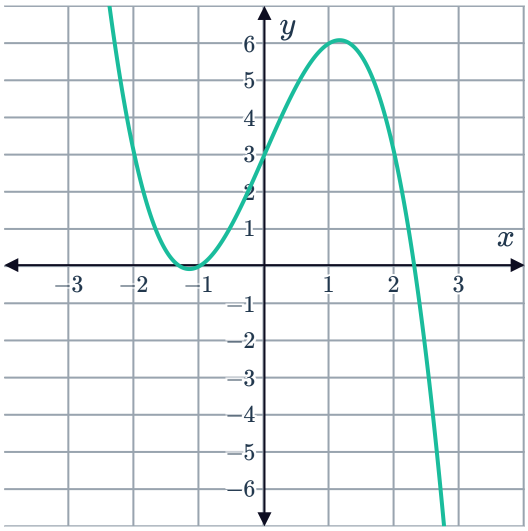


7 A cubic function is defined as $y = \dfrac{1}{2}x^3 + 4$.

a Find the x-intercept of the function.

b Find the y-intercept of the function.

8 Consider the graph of the function:



a The equation of the function can be written as $y = ax^3 + bx^2 + cx + d$.
Is the value of a is positive or negative?

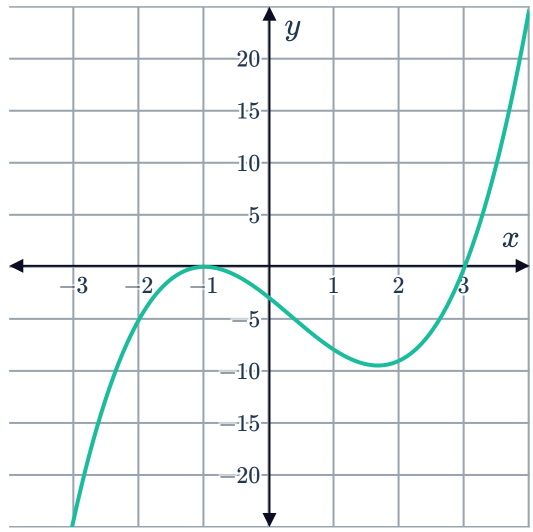
b State the coordinates of the y-intercept.

c For which values of x is the graph concave up?

d For which values of x is the graph concave down?

e State the coordinates of the point of inflection.

9 Consider the graph of the cubic function shown.
For what values of x is $y \geq 0$?



Transformations

10 Determine whether the following statements are true of the graphs of $y = \frac{1}{2}x^3$ and $y = x^3$.

a One is a reflection of the other about the y-axis.

b y increases more rapidly on $y = \frac{1}{2}x^3$ than on $y = x^3$.

c $y = \frac{1}{2}x^3$ is a horizontal shift of $y = x^3$.

d y increases more slowly on $y = \frac{1}{2}x^3$ than on $y = x^3$.

11 The graph of $y = x^3$ has a point of inflection at $(0, 0)$. By considering the transformations that have taken place, find the point of inflection of each cubic curve below:

a $y = \frac{2}{3}x^3$ b $y = x^3 + 3$ c $y = -x^3 + 4$

12 For each cubic function below:

i Complete a table of values of the form: ii Sketch the graph.

x	-2	-1	0	1	2
y					

a $y = x^3$ b $y = x^3 - 2$ c $y = -x^3 + 5$

13 Consider the curve $y = x^3 - 8$.

a Find the x-intercept.

b Find the y-intercept.

c Find the horizontal point of inflection.

d Sketch the graph of the curve.

14 Consider the curve $y = 3x^3 + 3$.

a Find the x-intercept.

b Find the y-intercept.

c State the coordinates of the point of inflection.

d Sketch the graph of the curve.

15 Consider the equation $y = x^3 - 3$.

a Complete the following set of points for the given equation.
$A(-3, \square), B(-2, \square), C(-1, \square), D(0, \square), E(1, \square), F(2, \square), G(3, \square)$

b Sketch the curve that results from the entire set of solutions for the equation being graphed.

16 Consider the cubic function $y = x^3 - 4$.

a State the y-intercept of the function.

b Complete the following table of values.

x	-1	1	2
y			

c Find the domain of the function in interval notation.

d Find the range of the function in interval notation.

e Sketch the graph.

17 Bianca completed the table of values for the equation $y = 20 - x^3$.

x	-6	-4	-2	0	2	4
y	236	84	28	20	-4	-44

a One of the points in the table is incorrect. Which point is it?

b Bianca wants to find one other pair of values that satisfy $y = 20 - x^3$ before graphing the curve. Find the ordered pair when the x-coordinate is 6.

c Plot the complete set of solutions for $y = 20 - x^3$, making sure that the curve goes through all points that satisfy it.

18 Consider the curve $y=-2x^3+16$.

a Find the x-intercept.

b Find the y-intercept.

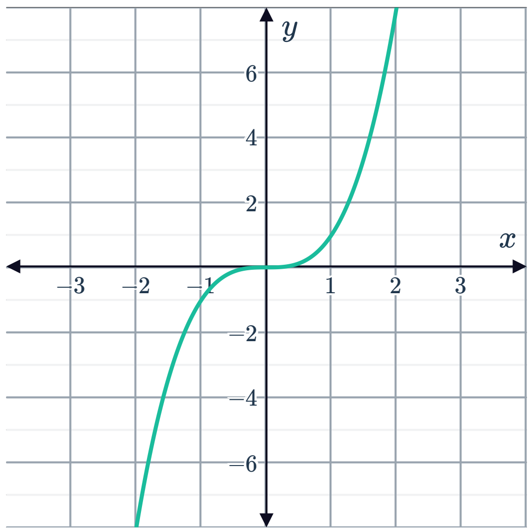
c State the coordinates of the point of inflection.

d Find the domain of the function in interval notation.

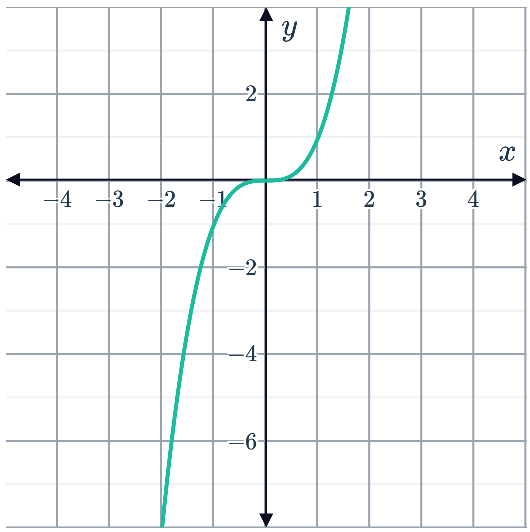
e Find the range of the function in interval notation.

f Sketch the graph of the curve.

19 A graph of $f(x)=x^3$ is shown. Sketch the curve after it has undergone transformations resulting in the function $g(x)=x^3-4$.



20 A graph of $y=x^3$ is shown. Sketch the curve after it has undergone transformations resulting in the function $y=2(x-2)^3-2$.



21 For the following cubic functions:

i Determine whether the cubic is increasing or decreasing from left to right.

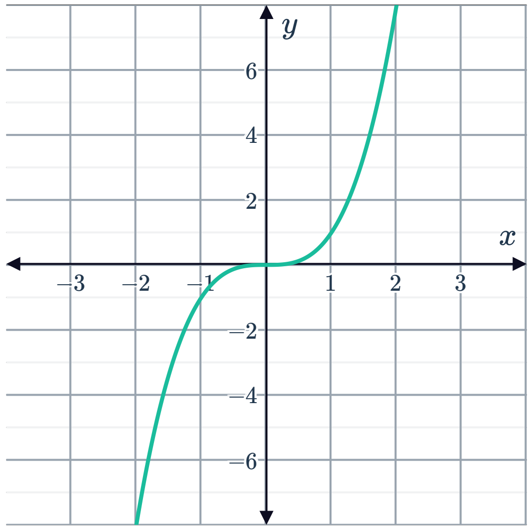
ii Determine whether the cubic is more or less steep than the cubic $y = x^3$.

iii Find the coordinates of the point of inflection of the cubic.

iv Sketch the graph.

a $y = 2x^3 + 2$ b $y = 4x^3 - 3$ c $y = -\frac{x^3}{4} + 2$

22 Consider the graph of $y = x^3$ shown:



a How do we shift the graph of $y = x^3$ to get the graph of $y = (x-2)^3 - 3$?

b Hence, sketch $y = (x-2)^3 - 3$.

23 Consider the function $y = 2(x-2)^3 - 2$

a Is the cubic increasing or decreasing from left to right?

b Is the function more or less steep than the function $y = x^3$?

c What are the coordinates of the inflection point of the function?

d Sketch the graph $y = 2(x-2)^3 - 2$.

24 Consider the curve $y = 2(x+1)^3 + 16$.

a Find the x-intercept.

b Find the y-intercept.

c State the coordinates of the point of inflection.

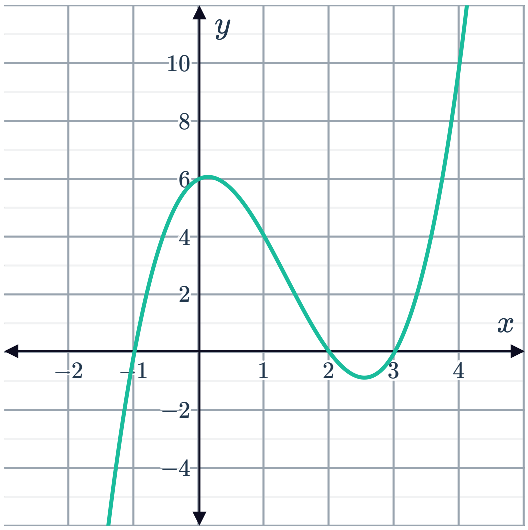
d Sketch the graph of the curve.

25 Consider the curve $y = -3(x-1)^3 + 3$.

- **a** Find the x-intercept.
- **b** Solve for the y-intercept.
- **c** State the coordinates of the point of inflection.
- **d** Sketch the graph of the curve.

Factored form of a cubic function

26 The cubic function $y = (x+1)(x-2)(x-3)$ has been graphed below. Determine the number of solutions to the equation $(x+1)(x-2)(x-3) = 0$.



27 A cubic function has the equation $y = x(x-4)(x-3)$. How many x-intercepts will it have?

28 State whether the following functions pass through the origin:

- **a** $y = (x-2)^2(x+3)$
- **b** $y = (x+1)^3$
- **c** $y = (x-4)(x+7)(x-5)$
- **d** $y = x(x-6)(x+8)$

29 State whether each function has exactly two x-intercepts:

- **a** $y = (x+3)^3$
- **b** $y = (x+6)^2(x+5)$
- **c** $y = (x+7)(x-1)(x-4)$
- **d** $y = x(x-2)(x-8)$

30 Consider the cubic function $y = (x+3)(x-2)(x-5)$.

- **a** Determine the x-intercepts.
- **b** A second cubic function has the same x-intercepts, but is a reflection of the above function about the x-axis. State the equation of the reflected function.

31 Sketch the function $y = (x-2)(x+1)(x+4)$ showing the general shape of the curve and the x-intercepts.

32 Sketch the graph of $f(x) = (x-2)^2(x-4)$ and $g(x) = -(x-2)^2(x-4)$ on the same number plane.

33 Consider the function $y = (x+3)^3$.

a. Complete the following table of values:

x	-5	-4	-3	-2	-1
y					

b. Sketch the graph.

c. State the domain.

d. State the range.

34 Consider the function $y = (x-2)^3$.

a. Complete the following table of values:

x	0	1	2	3	4
y					

b. Sketch the graph.

35 Consider the function $y = (x+1)^3$.

a. Complete the following table of values:

x	-3	-2	-1	0	1
y					

b. Sketch the graph.

36 Consider the equation $y = (x-4)^3$.

a. Complete the set of solutions for the above equation.
$A(3, \square), B(2, \square), C(5, \square), D(4, \square), E(6, \square)$

b. Plot the points on a coordinate axes.

c. On the same axes, plot the curve that results from the entire set of solutions for the equation being graphed.

d. Consider $x = 6.17$. According to the points on the graph, between which two integer values should the corresponding y-value lie?

37 For each of the functions below:

i Find the x-intercepts.

ii Find the y-intercept.

iii Sketch the graph.

a $y=(x+3)(x+2)(x-2)$

b $y=-(x+4)(x+2)(x-1)$

c $y=(x-2)^2(x+5)$

38 For each of the functions below:

i Express the equation in factorised form.

ii Find the y-intercept of the graph.

iii Find the x-intercepts of the graph.

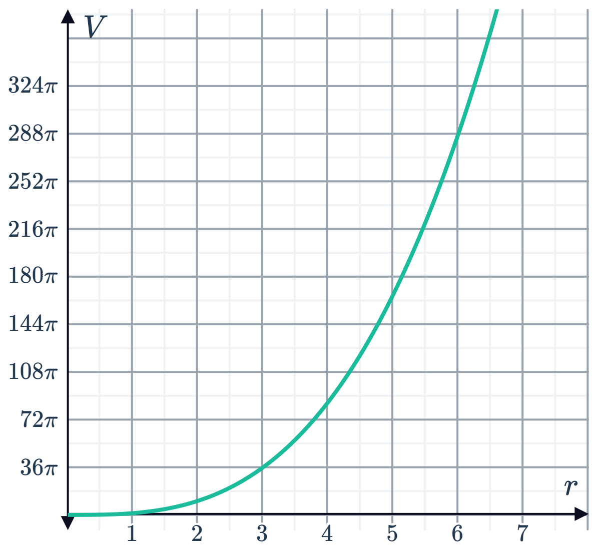
iv Sketch the graph of the curve.

a $y=3x+2x^2-x^3$

b $y=x^3-4x^2-7x+10$

Applications

39 The volume of a sphere has the formula $V=\frac{4}{3}\pi r^3$. The graph relating r and V is shown:



a Fill in the following table of values for the equation $V=\frac{4}{3}\pi r^3$, in terms of π:

r	1	2	4	5	7
V					

b Determine whether the following intervals show the volume, V of a sphere that has a radius measuring 4.5 m.

i $20\pi < V < 57\pi$

ii $85\pi < V < 166\pi$

iii $179\pi < V < 696\pi$

iv $221\pi < V < 366\pi$

c Using the graph, what is the radius of a sphere of volume $288\pi \text{ m}^3$?

40 A cube has side length 7 cm and a mass of [redacted] g. The mass of the cube is directly proportional to the cube of its side length.

a Let k be the constant of proportionality for the relationship between the side length x and the mass m of a cube. Find the value of k.

b Hence state the equation relating the mass (m) and side length (x) of a cube.

c Complete the table of values:

x	1	2	3	4
m				

d Sketch the graph of your equation.

e From the equation, find the mass of a cube with side 8.5 cm, to the nearest gram.

f A cube has a mass of 1920 g. From your graph, determine what whole number value its side length is closest to.

41 A box without a top cover is to be constructed from a rectangular cardboard, measuring 6 cm by 10 cm by cutting out four square corners of length $x \text{ cm}$. Let V represent the volume of the box.

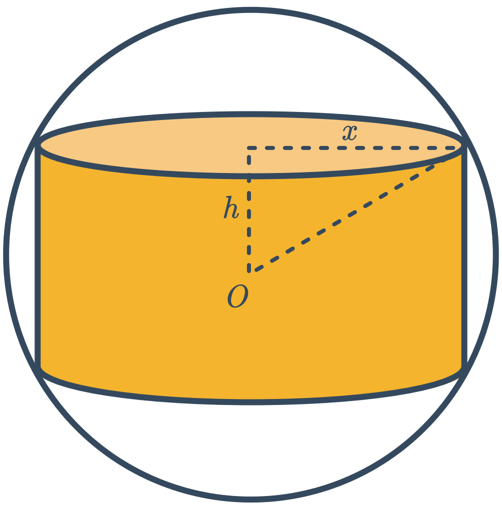
a Express the volume V of the box in terms of x, writing the equation in factorised form.

b For what range of values of x is the volume function defined?

c Sketch the graph of the volume function.

d Determine the volume of a box that has a height equivalent to the shorter dimension of the base.

42 A cylinder of radius x and height $2h$ is inscribed in a sphere of radius $\sqrt{15}$, centre at O as shown:



a Form an equation relating x and h.

b Form an expression for V, the volume of the cylinder, in terms of h.

c Determine the domain of h.

d The cylinder with the largest possible volume has a height of $2\sqrt{5}$, so $h = \sqrt{5}$. Determine the exact volume of this cylinder.